CRR Binomial Tree

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Overview

1. Binomial model review
2. Building the lattice
3. Backwards induction
4. Numerical aspects

References:

- "Options, Futures and Other Derivatives". J.C. Hull. Ed. Prentice Hall
- Cox, J., S. Ross and M. Rubinstein, 1979. Option Princing: A simplified approach. *Journal of Financial Economics* 7, 229-264.

One step binomial model

Consider a stock with current price $S_0 = 20$. At the end of three months, the stock price will be either 22 or 18. We are interested in the value of a call option with exercise price of 21.

$$S_0 = 20 \begin{array}{l} \nearrow 22\ (1) \\ \searrow 18\ (0) \end{array}$$

- We will **suppose** that no arbitrage opportunities exist.
- We set up a portfolio on the stock and the option without uncertainty (risk) at the end of the three months.
- This portfolio returns must equal the risk-free interest rate.

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- **Portfolio:** Long position in Δ shares of the stock and a short position in one call option.
- We calculate the value of Δ that makes the porfolio riskless

$$\begin{aligned} 22\Delta - 1 &= 18\Delta \\ &\downarrow \\ \Delta &= 0.25 \end{aligned}$$

Then, a riskless portfolio is a long position in 0.25 shares plus a short call.

At maturity (3 months)

- If $S_T = 22$, the value of the portfolio is $22 \times 0.25 - 1 = 4.5$
- If $S_T = 18$, the value of the portfolio is $18 \times 0.25 = 4.5$.

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Since the portfolio is riskless, in the absence of arbitrage opportunities, it must earn the risk-free of interest. Suppose $r = 12\%$. Then, the present value of the portfolio must be

$$4.5e^{-0.12\times 3/12} = 4.367$$

Let f denotes the option price and remember $S_0 = 20$. The value of the portfolio today is

$$20 \times 0.25 - f = 5 - f \longrightarrow 5 - f = 4.367 \Longrightarrow \mathbf{f = 0.633}$$

in the absence of arbitrage opportunities.

Note: We don't have use the probabilities of up and down movement to derive the option price.

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A Generalization

- S_0: The present of one share.
- $S_0 u$: Value of one share after an up movement $(u > 1)$. The proportional increase in the stock price is $u - 1$.
- $S_0 d$: Value of one share after a down movement $(d < 1)$. The proportional decrease in the stock price is $1 - d$.
- f_u, f_d: Value of the call option after the up or down movement respectively.

$$\begin{tikzcd}[row sep=small] & S_0 \times u \\ S_0 \arrow[ur] \arrow[dr] & \\ & S_0 \times d \end{tikzcd} \qquad \begin{tikzcd}[row sep=small] & f_u \\ f \arrow[ur] \arrow[dr] & \\ & f_d \end{tikzcd}$$

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$$\begin{array}{ccc} & \nearrow & S_0 \times u \\ S_0 & & \\ & \searrow & S_0 \times d \end{array} \qquad \begin{array}{ccc} & \nearrow & f_u \\ f & & \\ & \searrow & f_d \end{array}$$

A Generalization (II)

We consider a portfolio consisting of a long position in Δ shares and a short position in one option.

	up movement	down movement
Portfolio Value	$S_0 u\Delta - f_u$	$S_0 d\Delta - f_d$

$$S_0 u\Delta - f_u = S_0 d\Delta - f_d$$

$$\Downarrow$$

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

Δ is the ratio of the change in the option price to the change in the stock price.

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A Generalization (III)

- Cost of setting up the portfolio: $S_0\Delta - f$
- Present value of the portfolio: $(S_0 u\Delta - f_u)e^{-rT}$

$$S_0\Delta - f = (S_0 u\Delta - f_u)e^{-rT}$$

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$$f = S_0\Delta - (S_0 u\Delta - f_u)e^{-rT}$$

If we replace Δ by $\frac{f_u - f_d}{S_0 u - S_0 d}$

$$f = e^{-rT}\left[pf_u + (1-p)f_d\right]$$

where

$$p = \frac{e^{rT} - d}{u - d}$$

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Example

Suppose $u = 1.1,\ d = 0.9,\ r = 0.12,\ T = 0.25,\ f_u = 1,\ f_d = 0.$

$$p = \frac{e^{0.12\times 0.25} - 0.9}{1.1 - 0.9} = 0.6523$$

$$\Downarrow$$

$$f = e^{-0.12\times 0.25}\left[0.6523 \times 1 + 0.3477 \times 0\right] = 0.633$$

Notice that we would get the same option price when the probability of an upward movement is 0.5 as we do when it is 0.9.

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Risk-Neutral Valuation

$$f = e^{-rT}\left[pf_u + (1-p)f_d\right]$$

Although we do not need to make any assumptions about up and down probabilities, it is natural to interpret p $(1-p)$ as the probability of an up (down) movement. Then

$$pf_u + (1-p)f_d$$

is the expected payoff from the option.

Thus, the first equation states that the value of the option today is its expected future value discounted at the risk-free rate, $\Longrightarrow$

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The Cox-Ross-Rubinstein Binomial Model

Model parameters

- Time to maturity: T years.
- nt time steps $\to \delta t = \frac{T}{nt}$ is the time length of one time step.
- r and q are the yearly continuously compounded risk free interest rate and dividend yield, respectively.
- To match the volatility, σ and the expected return:
  - ♣ $u = e^{\sigma\sqrt{\delta t}}$, $\quad d = \frac{1}{u} = e^{-\sigma\sqrt{\delta t}}$
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Multi-step binomial model (matrix form)

$$\begin{bmatrix} S_0 & S_0u & S_0u^2 & S_0u^3 & S_0u^4 \\ & S_0d & S_0 & S_0u & S_0u^2 \\ & & S_0d^2 & S_0d & S_0 \\ & & & S_0d^3 & S_0d^2 \\ & & & & S_0d^2 \end{bmatrix} = S_0 \times \begin{bmatrix} u^0 & u^1 & u^2 & u^3 & u^4 \\ & u^{-1} & u^0 & u^1 & u^2 \\ & & u^{-2} & u^{-1} & u^0 \\ & & & u^{-3} & u^{-2} \\ & & & & u^{-4} \end{bmatrix} =$$

$$= S_0 \times u^J, \quad \text{where} \quad J = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ & -1 & 0 & 1 & 2 \\ & & -2 & -1 & 0 \\ & & & -3 & -2 \\ & & & & -4 \end{bmatrix}$$

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- row 4: $J(4, 4 : end) = J(3, 3 : 4) - 1$;
- row 5: $J(5, 5 : end) = J(4, 4 : 4) - 1$;

A general expression for nt time-steps

- row 1: $J(1, 1 : end) = (0 : nt)$;
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A general expression for nt time-steps

- row 1: $J(1, 1 : end) = (0 : nt)$;
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Multi-step binomial model (matrix form)

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Example

Suppose $S_0 = 30,\ r = 0.05,\ q = 0,\ T = 1,\ \sigma = 0.30,\ nt = 4.$

$$S = \begin{bmatrix} 30 & 34.855 & 40.496 & 47.049 & 54.664 \\ & 25.821 & 30 & 34.855 & 40.496 \\ & & 22.225 & 25.821 & 30 \\ & & & 19.129 & 22.225 \\ & & & & 16.464 \end{bmatrix}$$

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Objective:

To build two binomial trees (matrix) for the value of the derivatives, f^c and f^p (call and put), using the underlying matrix, S.

The procedure:

How to do it? By backwards induction: from maturity, T, to the current date. The arbitrage-free value of the option is the expected value computed using the risk-neutral probabilities.

At maturity:

The option can only be exercised if it is in the money.

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Suppose $K = 31$, then at maturity

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$$f_T^c = \begin{bmatrix} - & - & - & - & 23.66 \\ - & - & - & - & 9.49 \\ - & - & - & - & 0 \\ - & - & - & - & 0 \\ - & - & - & - & 0 \end{bmatrix}; \quad f_T^p = \begin{bmatrix} - & - & - & - & 0 \\ - & - & - & - & 0 \\ - & - & - & - & 1 \\ - & - & - & - & 8.77 \\ - & - & - & - & 14.54 \end{bmatrix}$$

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At intermidiate time steps $t < T$, the option will be exercised if it is more profitable than continuing.

Exercise value:

$$exe^{c}_{i,t} = max(S_{i,t} - K, 0); \qquad exe^{p}_{t} = max(K - S_{i,t}, 0)$$

Expected value using the risk neutral probability (p):

$$E^{Q}_{t}[f_{i,t+1}] = e^{-r\delta t}\left[p \times f_{i,t+1} + (1-p) \times f_{i-1,t+1}\right]$$

Value of the derivative:

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At $T-1$

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Continuation value at $T-1$

Expected value using the risk neutral probability (p):

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Optimal Strategy

Value of the derivative:

$$f_{i,t} = max\left(E_t^Q[f_{i,t+1}],\ exe_{i,t}^{c,p}\right)$$

$$f_{T-1}^p = max\begin{pmatrix} 0.00 & , & 0.00 \\ 0.00 & , & 0.49 \\ 5.18 & , & 4.79 \\ 11.87 & , & 11.49 \end{pmatrix} \Rightarrow \quad f_{T-1}^p = \begin{pmatrix} \dots & 0.00 & 0 \\ \dots & 0.49 & 0 \\ \dots & 5.18 & 1 \\ \dots & 11.87 & 8.77 \\ \dots & - & 14.54 \end{pmatrix}$$

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The derivatives tree

$$f^c = \begin{bmatrix} \mathbf{3.75} & 6.38 & 10.50 & 16.43 & 23.66 \\ & 1.17 & 2.36 & 4.73 & 9.49 \\ & & 0 & 0 & 0 \\ & & & 0 & 0 \\ & & & & 0 \end{bmatrix}$$

$$f^p = \begin{bmatrix} \mathbf{3.52} & 1.48 & 0.24 & 0 & 0 \\ & 5.68 & 2.78 & 0.49 & 0 \\ & & 8.78 & 5.18 & 1 \\ & & & 11.87 & 8.78 \\ & & & & 14.54 \end{bmatrix}$$

The number of time steps, *nt*

The larger the number of time steps, the better the approximation.

Nonetheless, the larger the number of time steps, also the larger the computational time.

How behaves the derivative price with the number of time steps?

Exercise: Show how behaves the call and put prices with the number of time steps.

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